gpeyt1-05 · (N1)60 profile

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sand deposit has γ = 118 pcf above the water table at 6 ft and γsat = 128 pcf below. Two SPT samples were taken with a safety hammer (ER = 70%): N = 11 at 12 ft and N = 24 at 45 ft. The geologist claims the deeper sand is 'much denser' because its raw N is ~65% higher. The rig used 2 in. OD split spoon and 1.4 in. ID liner-free sampler.

Most nearly, (N1)60 at 45 ft is:

A. 28
B. 18
C. 23
D. 15

Given equations / data

> N60 = (ER/60)·N; CN = 0.77 log10(20/Po), Po in tsf effective; γw = 62.4 pcf

Worked solution

1. Po(12) = 0.551 tsf → (N1)60 = 11×70/60×1.201 = 15.4
2. Po(45) = [6(118) + 39(128−62.4)]/2000 = 1.633 tsf
3. CN = 0.77 log10(20/1.633) = 0.838
4. (N1)60 = 24×70/60×CN = 23.5; after correction the two sands are much closer than raw N suggests

Final answer

23

Why the other values are wrong

- **28** — forgot the overburden correction
- **18** — used total stress in CN
- **15** — reported the 12 ft value

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2, p. 124

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwQ24-zlxyU (no creator values copied).

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